

Standard Form & Indices

Test 8 · 41 marks total · Sections F & G include Higher-tier content

Name: _____

Date: _____

Score: _____ / 41

Section A — Laws of indices

A1. Write $6^4 \times 6^7$ as a single power of 6.

[1 mark]

A2. Write $2^{15} \div 2^9$ as a single power of 2.

[1 mark]

A3. Write $(9^3)^4$ as a single power of 9.

[1 mark]

A4. Work out the value of $3^3 \times 3^1$.

[2 marks]

Section B — Zero and negative indices

B1. Work out the value of 41^0 .

[1 mark]

B2. Work out the value of 8^{-2} . Give your answer as a fraction.

[2 marks]

B3. Work out the value of 5^{-1} . Give your answer as a decimal.

[2 marks]

Section C — Writing standard form

C1. Write 63,000 in standard form.

[2 marks]

C2. Write 1,700,000 in standard form.

[2 marks]

C3. Write 0.0029 in standard form.

[2 marks]

C4. Write 0.000004 in standard form.

[2 marks]

Section D — Converting back, and calculating

D1. Write 2.6×10^5 as an ordinary number.

[1 mark]

D2. Write 8×10^{-4} as an ordinary number.

[1 mark]

D3. Work out $(6 \times 10^5) \times (7 \times 10^2)$. Give your answer in standard form.

[3 marks]

D4. Work out $(2 \times 10^9) \div (8 \times 10^3)$. Give your answer in standard form.

[3 marks]

Section E — Powers to know, and estimating roots

E1. Write down the value of 4^3 .

[1 mark]

E2. $\sqrt{20}$ lies between which two consecutive whole numbers?

[1 mark]

Section F — Laws of indices with algebra

F1. Simplify $g^5 \times g^9$.

[1 mark]

F2. Simplify $h^{13} \div h^4$.

[1 mark]

F3. Simplify $(a^7)^2$.

[1 mark]

F4. Simplify $7b^3 \times 3b^5$.

[2 marks]

F5. Simplify $(3c^4)^2$.

[2 marks]

Section G — Fractional indices (Higher)

G1. Work out the value of $121^{1/2}$.

[1 mark]

G2. Work out the value of $512^{1/3}$.

[1 mark]

G3. Work out the value of $4^{5/2}$.

[2 marks]

G4. Work out the value of $27^{-1/3}$.

[2 marks]